

TCMC • HARMONY EARS

Chord Progression

Pitch Trainer

User Manual

(和弦進行音準練習)

App Version: v1.0

Producer

Taiwan Choral Music Center (TCMC)

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1. Overview

"Chord Progression Pitch Trainer" is a web-based tool from the Taiwan Choral Music Center (TCMC) that lets you pick a common chord progression (such as I-IV-V-I, "6415," or "16251"), a key, and the voice part you want to practice (Soprano/Alto/Tenor/Bass), then sing your way through the whole progression along with a metronome while a real-time microphone check tracks your pitch.

This app uses the same tuning approach as the companion "Chord Voicing Trainer": each chord's root sits at standard 12-tone equal temperament, while its third, fifth and seventh are tuned in Just Intonation (pure ratios). The goal is to build a feel for the "locked-in" sound of in-tune chords, not just hitting the right note names.

Key Features

- 16 built-in chord progressions, covering common major/minor progressions, dominant-7th/turnaround (vi7-ii7-V7) variants, and a "Golden Progression" that includes chord inversions and passing 7th chords
- Every progression can be sung in all 12 keys (all 12 have been checked and adjusted)
- Once a voice part is selected, each part has its own / toggle so you can choose which parts you hear — mute your own part once you know it
- An on/off metronome, adjustable tempo (40–160 BPM), and a loop toggle to repeat the whole progression
- A  "Voicing Table" showing, for every chord in the progression, each voice part's solfège syllable, note name, and offset from equal temperament in cents — tap any cell to hear that note
- Real-time microphone pitch detection with a ± 6 cent tolerance band, shown live on the chart
- Chinese / English interface toggle

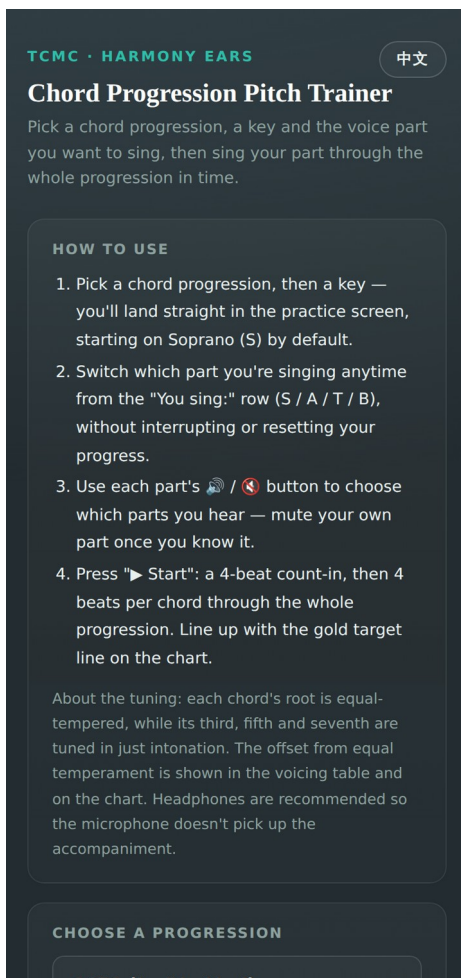


Tip: Headphones are recommended. Over speakers, the accompaniment can bleed into the mic and throw off the pitch judgement.

2. Interface Overview

Screen 1: Choose a Progression

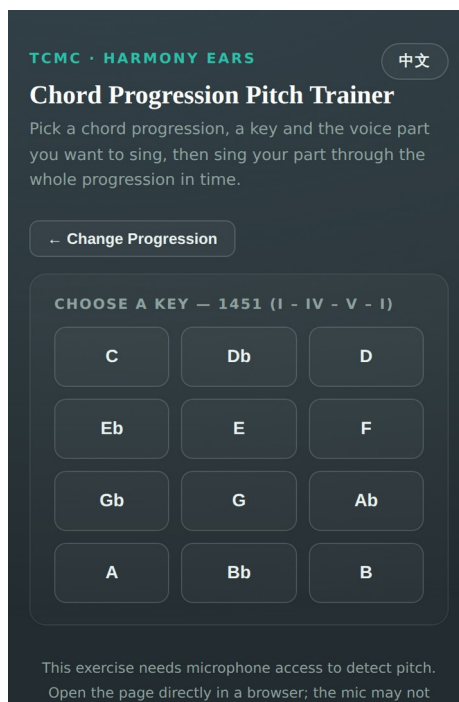
A short "How to Use" summary appears at the top, followed by a list of the 16 built-in chord progressions. Tap one to continue.



Screen 1: Choose a Progression

Screen 2: Choose a Key

The 12 keys are shown as buttons — all 12 are available to choose from. Once you pick one you land straight in the practice screen — there's no separate voice-part step, since the practice screen's "You sing:" row lets you switch parts anytime.



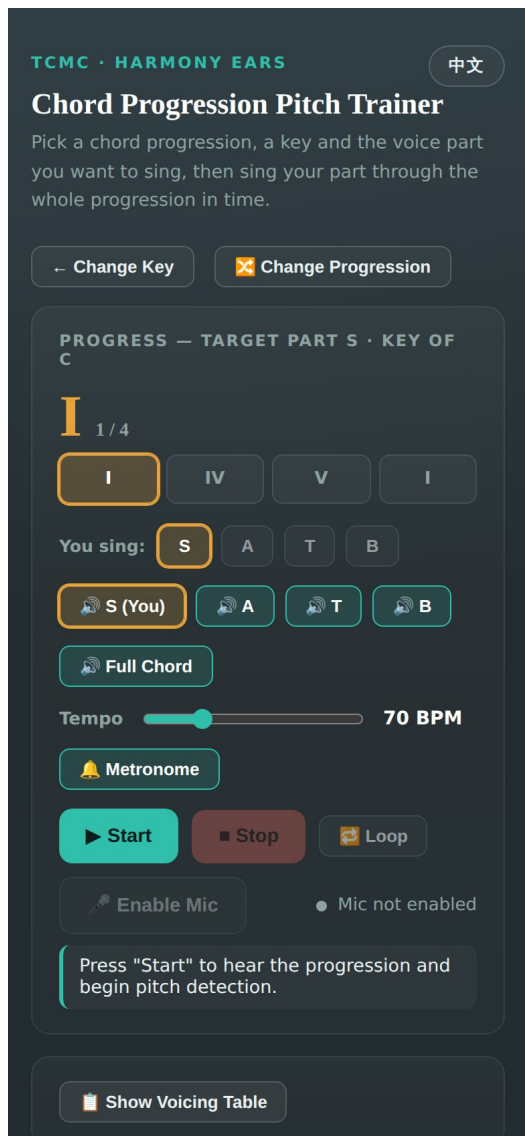
Screen 2: Choose a Key

Screen 3: Practice Screen

This is the main practice screen — you land here right after choosing a key, starting on Soprano (S) by default. From top to bottom it contains:

- A progress row — every chord in the progression (roman numerals), with the current one highlighted in gold; press and hold any chord box to hear that chord, and release to stop
- A "You sing:" row — tap to switch your target voice part (S / A / T / B) at any time, without interrupting or resetting your progress
- Per-part / toggles plus a "Full Chord" master toggle, controlling which parts you hear during playback
- A tempo slider and metronome toggle
- Start / Stop / Loop / mic toggle
- A Voicing Table (collapsed by default — tap to expand)
- The pitch chart with live readouts (current pitch, offset from target, judgement)

Tip: Right next to the " Enable Mic" button, the screen always shows a reminder: if it's not picking up sound, check your browser's microphone permission (the icon in the address bar) and make sure the right input device is selected — Bluetooth headphones and external mics are the most common cause of picking the wrong one.



Screen 3: Practice Screen

3. How to Use

1. From the first screen, pick the chord progression you want to practice.
2. Choose a key — you'll land straight in the practice screen, starting on Soprano (S) by default.
3. Tap the part you want to sing in the "You sing:" row (S / A / T / B). You can switch anytime — it never interrupts or resets your progress.
4. (Optional) Open the  Voicing Table first to preview every chord's solfège and note names for your part — tap any cell to hear it.
5. Use each part's  /  toggle to decide which parts you hear. While you're still learning it, leave everything on; once you know your part, mute it and rely on your ear and the pitch chart instead.
6. Adjust the tempo (BPM) and metronome toggle if needed.
7. Press " Start": a 4-beat count-in plays, then each chord plays for 4 beats (speed set by the tempo slider) through the whole progression. Your browser will prompt for microphone access — choose Allow.
8. Sing along, aligning with the gold target line on the pitch chart. Your trace shows green within ± 6 cents of the target, and red outside that band.
9. The session stops automatically once the progression finishes; with " Loop" enabled it restarts from the top automatically. You can also press "■ Stop" at any time.
10. To change your key or progression, use the "← Change Key" or " Change Progression" buttons in the top-left corner — no page reload needed.

 **Tip:** If you're using an in-chat or in-app preview window, browser security restrictions may block the microphone permission request — download this HTML file and open it directly in a browser instead.

4. Progressions at a Glance

There are currently 16 built-in progressions:

Progression	Description
I - IV - V - I (1-4-5-1)	The most basic major-key cadence progression
vi - IV - I - V (6m-4-1-5)	A common pop-song progression
I - vi - ii - V - I (1-6m-2m-5-1)	A common jazz/pop progression including the ii chord
Cycle of Thirds (I - vi - IV - ii - vii° - V - iii - I) (1-6m-4-2m-7°-5-3m-1)	A descending cycle built on thirds
I - IV - I (1-4-1)	Basic progression
I - V - I (1-5-1)	Basic progression
I - V - IV - I (1-5-4-1)	Basic progression
I - vi - I (1-6m-1)	Basic progression
I - iii - IV - I (1-3m-4-1)	Basic progression
i - iv - V - i (minor) (1m-4m-5-1m)	Natural-minor cadence with a major dominant
I - IV - I - V - I (1-4-1-5-1)	Basic progression
i - VI - VII - i (natural minor) (1m-6-7-1m)	A natural-minor progression
I - IV - V7 - I (1-4-57-1)	The dominant-7th version of 1451
i - iv - V7 - i (minor) (1m-4m-57-1m)	A minor-key dominant-7th cadence
I - vi7 - ii7 - V7 - I (1-6m7-2m7-57-1)	A jazz-style "turnaround" built with 7th chords
Golden Progression (I - V6 - vi - iii6(7) - IV - I6 - ii(7) - V) (1-5(6)-6m-3m(6)7-4-1(6)-2m7-5)	Features chord inversions and passing 7th chords, prioritizing smooth stepwise voice leading

 **Tip:** The number in parentheses is a shorthand scale-degree label: minor chords get an "m" (e.g. 6m, 2m), the diminished chord gets a "°" (e.g. 7°), 7th chords get a "7" straight after the number (e.g. 57, 2m7), and inverted chords get "(6)" (e.g. 5(6), 1(6)).

 **Tip:** All 16 progressions have been verified to be singable in all 12 keys, with every voice part's notes falling within a comfortable range.

5. Voicing Table

Tap "📄 Show Voicing Table" to expand a table: columns are each chord in the progression (labeled with roman numerals), rows are the S / A / T / B voice parts, and your currently selected target part is marked with a ★.

Each cell shows:

- Large text: the solfège syllable (movable-do — Do is the tonic in a major key, La is the tonic in a minor key)
- Smaller text below: the note name (e.g. C4, Eb4)
- Top-right number: which chord tone this is (1 root, 3 third, 5 fifth, 6 sixth, 7 seventh)
- Below that: the offset from equal temperament in cents (e.g. +2¢, -14¢), so you can check the tuning direction before you sing

Press and hold any cell to hear that exact note for as long as you hold it — handy for checking what you're about to sing before pressing Start. Press and hold a roman numeral in the top row to hear the whole chord at once.

📄 Hide Voicing Table

	I	IV	V	I
S ★	Sol ⁵ +2¢ G4	La ³ -14¢ A4	Sol ¹ +0¢ G4	Sol ⁵ +2¢ G4
A	Mi ³ -14¢ E4	Fa ¹ +0¢ F4	Re ⁵ +2¢ D4	Mi ³ -14¢ E4
T	Do ¹ +0¢ C4	Do ⁵ +2¢ C4	Ti ³ -14¢ B3	Do ¹ +0¢ C4
B	Do ¹ +0¢ C3	Fa ¹ +0¢ F2	Sol ¹ -0¢ G2	Do ¹ +0¢ C3

In each cell: solfège syllable (large) and note name. Top right: which chord tone it is (1 root, 3 third, 5 fifth, 6 sixth, 7 seventh) and, below it, the offset from equal temperament in cents. Tap any

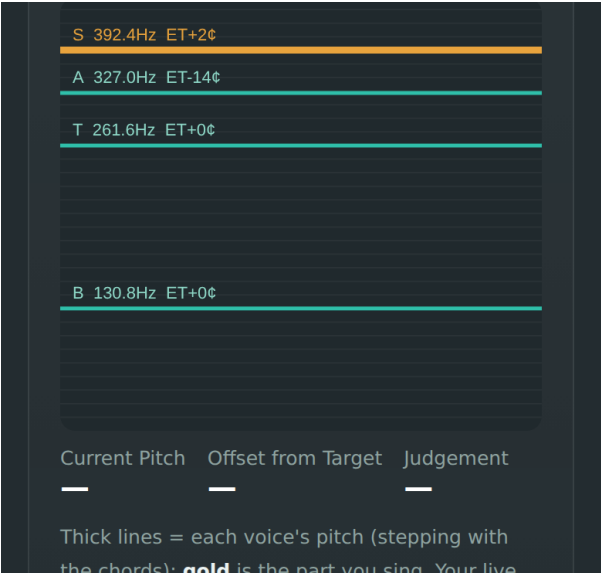
Voicing table example

6. Pitch Chart & Tuning Judgement

The chart's vertical range is fixed across the whole progression, so the reference lines visibly move up and down as chords change, rather than re-centering (and looking static) every time the chord changes.

The thick stepped lines mark each voice part's reference pitch; gold marks your current singing target. Once you start singing, your live pitch trace appears in green when within ± 6 cents of the target, and red when outside that band.

Below the chart you'll also see the detected frequency (Hz), the offset in cents from the target, and a text judgement (✓ In tune / Sharp / Flat).



Pitch chart example (before detection starts)

7. Troubleshooting

Microphone not responding / can't detect pitch

- Confirm your browser has allowed this page to use the microphone (check the permission icon in the address bar)
- If you're using an in-chat or in-app preview window, security restrictions may block the microphone — download the HTML file and open it directly in a browser instead
- Confirm the URL starts with HTTPS; a plain local file (file://) often can't get microphone access on some mobile browsers

No sound

- Check that your device isn't muted and the browser tab hasn't been silenced by the system
- Some mobile browsers need one manual tap before they'll unlock audio playback — if there's no sound the first time, try tapping again

Pitch trace jumps around or the judgement feels unstable

- Headphones are recommended over speakers, to avoid the accompaniment bleeding into the mic and throwing off detection
- Sing in a quiet environment and keep a consistent distance from the microphone

8. Version History

Current version: v1.0

Development Highlights

- v1.0: initial public release. All 16 chord progressions verified singable in all 12 keys; bilingual interface, voicing table, metronome and tempo control, and real-time microphone pitch detection.

Production Team

Produced by: Taiwan Choral Music Center (TCMC)

Producer: Ray Chu (朱元雷)

Nature of software: a web-based tool for choral voice-part rehearsal and self-practice